Module 2: Foundations in biology

All living organisms have similarities in cellular structure, biochemistry and function. An understanding of these similarities is fundamental to the study of the subject.

This module gives learners the opportunity to use microscopy to study the cell structure of a variety of organisms. Biologically important molecules such as carbohydrates, proteins, water and nucleic acids are studied with respect to their structure and function. The structure and mode of action of enzymes in catalysing biochemical reactions is studied.

Membranes form barriers within, and at the surface of, cells. This module also considers the way in which the structure of membranes relates to the different methods by which molecules enter and leave cells and organelles.

The division and subsequent specialisation of cells is studied, together with the potential for the therapeutic use of stem cells.

Learners are expected to apply knowledge, understanding and other skills developed in this module to new situations and/or to solve related problems**.**

2.1 Foundations in biology

2.1.1 Cell structure

Biology is the study of living organisms. Every living organism is made up of one or more cells, therefore understanding the structure and function of the cell is a fundamental concept in the study of biology. Since Robert Hooke coined the phrase 'cells' in 1665, careful observation using microscopes has revealed details of cell structure and ultrastructure and provided evidence to support hypotheses regarding the roles of cells and their organelles.

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the use of microscopy to observe and investigate different types of cell and cell structure in a range of eukaryotic organisms	To include an appreciation of the images produced by a range of microscopes; light microscope, transmission electron microscope and scanning electron microscope. HSW1, HSW7
(b)	the preparation and examination of microscope slides for use in light microscopy	Including the use of an eye piece graticule and stage micrometer. **PAG1** HSW4
(c)	the use of staining in light microscopy	To include the use of differential staining to identify different cellular components and cell types. **PAG1** HSW4, HSW5
(d)	the representation of cell structure as seen under the light microscope using drawings and annotated diagrams of whole cells or cells in sections of tissue	**PAG1**

(e)	the use and manipulation of the magnification formula	$\text{magnification} = \dfrac{\text{size of image}}{\text{size of real object}}$ *M0.1, M0.2, M0.3, M1.1, M1.8, M2.2, M2.3, M2.4*
(f)	the difference between magnification and resolution	To include an appreciation of the differences in resolution and magnification that can be achieved by a light microscope, a transmission electron microscope and a scanning electron microscope. Learners are **not** required to recall exact resolutions or magnification numbers for each of the listed microscopes. *M0.2, M0.3* HSW7, HSW8
(g)	the ultrastructure of eukaryotic cells and the functions of the different cellular components	To include the following cellular components and an outline of their functions: nucleus, nucleolus, nuclear envelope, rough and smooth endoplasmic reticulum (ER), Golgi apparatus, ribosomes, mitochondria, lysosomes, chloroplasts, plasma membrane, centrioles, cell wall, flagella and cilia. *M0.2*
(h)	photomicrographs of cellular components in a range of eukaryotic cells	To include interpretation of transmission and scanning electron microscope images.
(i)	the interrelationship between the organelles involved in the production and secretion of proteins	
(j)	the importance of the cytoskeleton	To include providing mechanical strength to cells, aiding transport within cells and enabling cell movement. HSW2
(k)	the similarities and differences in the structure and ultrastructure of prokaryotic and eukaryotic cells.	**PAG1**